DeFi & Advanced Crypto

Module 1 Study Guide

DeFi Ecosystem Overview

This guide covers every lesson of the module, using only words and figures from the lessons themselves. Work through it after watching the lessons, then use the checklist at the end to test yourself.

Lesson 1: What DeFi Actually Is (and Isn't)

In this lesson: Centralized Banking & Exchanges; Decentralized Smart Contracts; Custody & Ris

Key points

- Centralized models hold custody and control your transaction flow
- DeFi uses smart contracts to keep you in direct self-custody
- Technical and code risks shift completely to the individual user

Examples from the lesson

- We are stepping into our course on DeFi and Advanced Crypto, beginning with Module 1, "DeFi Ecosystem Overview," and our very first lesson, "What DeFi Actually Is (and Isn't)."
- Today, we are focusing on one fundamental objective: distinguishing decentralized protocols from centralized crypto exchanges and traditional banks so you always know where control, custody, and risk sit in each model.
- Imagine you decide to provide liquidity using TT$6,000 worth of digital assets to earn a share of network transaction fees.
- You retain self-custody of your private keys in your personal wallet while depositing that TT$6,000 directly into an automated smart contract.
- In this scenario, your initial deposit sits as TT$6,000 in smart contract liquidity.
- Over time, as users trade through the pool, the protocol automatically collects and allocates TT$300 in trading fees directly to your pool balance based on coded rules.
- That brings your total asset balance in the smart contract to TT$6,300.
- Because control remains programmatic rather than manual, you can trigger a withdrawal transaction through your private key whenever you choose, pulling your full TT$6,300 back into your wallet without asking permission from a manager or waiting for bank opening hours.

Lesson 2: Mapping the DeFi Stack

In this lesson: Core Building Blocks; Stablecoins & DEXs; Lending Markets & Aggregators; Trac

Key points

- Four core layers form the DeFi stack
- Transactions flow across settlement, lending, and DEXs
- Aggregators automate execution across protocols

Examples from the lesson

- I am glad you are joining us for Course 4, DeFi and Advanced Crypto.
- We are still in Module 1, DeFi Ecosystem Overview, and today we are looking at Lesson 2, Mapping the DeFi Stack.
- A DEX is like a peer-to-peer market stall where you swap one token for another directly from your personal digital wallet.
- Instead of checking four different places yourself, the aggregator routes your transaction across the stack in a single click.
- Imagine you start with TT$6,800, which converts to roughly 1,000 USDC on the blockchain.
- You deposit that 1,000 USDC into a lending pool as collateral so you can borrow against it without selling your baseline capital.
- The lending protocol allows you to borrow 600 USDC worth of another stablecoin, say DAI, against your deposit.
- You then take that 600 DAI to a DEX aggregator to swap it for a different growth token.
- The aggregator routes 300 DAI through DEX A and 300 DAI through DEX B to execute your trade with the lowest slippage possible.
- Notice how a single strategy moved across three distinct layers: stablecoins provided the settlement, the lending market provided the leverage, and the DEX aggregator executed the swap.
- The DeFi stack is built on four core layers: stablecoins for stable value, DEXs for token trading, lending markets for borrowing and earning, and aggregators for routing efficiency.
- In our next lesson, Lesson 3, Reading a Protocol Before You Touch It, we will learn how to inspect smart contracts, evaluate liquidity pools, and spot red flags before committing a single dollar.

Lesson 3: Reading a Protocol Before You Touch It

In this lesson: Total Value Locked (TVL); Docs & Audits; Governance & Controls; Community Sig

Key points

- Check TVL stability over time to verify real usage
- Inspect third-party audits and public documentation thoroughly
- Require transparent governance and healthy developer community interactions

Examples from the lesson

- We are wrapping up Module 1, DeFi Ecosystem Overview, with Lesson 3: Reading a Protocol Before You Touch It.
- We use four main signals to build a shortlist worth researching further: Total Value Locked, documentation, governance, and community signals.
- Let's break down how to inspect each one step by step.
- If two founders can quietly sweep the treasury at midnight without community approval, your funds are at serious risk.
- Now, let's run through a practical evaluation exercise using a prospective TT$10,000 crypto asset test portfolio.
- You assign points across our four signals using a strict 100-point safety scorecard.
- You set a strict threshold rule: any protocol scoring below 70 points gets instantly discarded from your shortlist, saving you from high-risk landmines.
- For TVL, it has TT$5,000,000 in steady deposits over six months, earning a solid 25 points out of 25.
- For documentation and security, it has two clean public audits, scoring another 25 points out of 25.
- However, when you check governance, you find the founders hold 80% of all voting tokens, so it scores a weak 10 points out of 25.
- Finally, its community admins are evasive in Discord, earning just 10 points out of 25.
- When you add up these signal scores, Horizon Yield reaches a total score of 70 points out of 100.
- Let's compare that score directly against your minimum shortlist threshold of 70 points out of 100.
- Evaluating a DeFi protocol requires examining four core pillars: Total Value Locked, clear audited documentation, transparent governance, and healthy community signals.

Recap checklist — can you say each one?

Lesson 1: What DeFi Actually Is (and Isn't)

- ☐ Centralized models hold custody and control your transaction flow
- ☐ DeFi uses smart contracts to keep you in direct self-custody
- ☐ Technical and code risks shift completely to the individual user

Lesson 2: Mapping the DeFi Stack

- ☐ Four core layers form the DeFi stack
- ☐ Transactions flow across settlement, lending, and DEXs
- ☐ Aggregators automate execution across protocols

Lesson 3: Reading a Protocol Before You Touch It

- ☐ Check TVL stability over time to verify real usage
- ☐ Inspect third-party audits and public documentation thoroughly
- ☐ Require transparent governance and healthy developer community interactions